

GREAT SCIENTISTS

Louis Pasteur

French chemist Louis Pasteur (1822–1895) made several important contributions to science—from introducing pasteurization (the process of briefly heating a substance to kill microbes) to developing vaccines that could fight deadly diseases.

Germ theory

The idea that infectious diseases are spread by germs, or microorganisms, is known as germ theory. Today, we take it for granted, but this view was highly controversial when Pasteur, then a Professor of Chemistry, began his work in the 1860s. He showed that tiny organisms were responsible for contaminating beer and milk. He also found they caused diseases in silkworms.

Work on vaccines

Pasteur's confirmation of germ theory revolutionized medicine and healthcare by introducing new standards of hygiene. He developed vaccines against two animal diseases—chicken cholera and anthrax—by producing weakened strains of the disease-carrying bacteria in the laboratory. Once injected into animals, these strains produced antibodies (proteins the immune system creates to attack unfamiliar substances) that gave the animals immunity against the disease.

Fight against rabies

Pasteur made a vaccine against rabies, a disease spread to humans by infected animals. In 1886, he used it to successfully cure nine-year-old Joseph Meister, who had been bitten by an infected dog.

Dedication to science

In 1868, Pasteur had a stroke that partially paralyzed the left side of his body. He continued working on his research, with the help of colleagues and assistants. He died in 1895.



The Pasteur Institute

A doctor vaccinates a patient against rabies in the Pasteur Institute, the center founded by Pasteur in Paris, France, for the study of diseases and vaccines. It still carries out life-saving research in all parts of the world.



Silkworm research

Shown above is the microscope Pasteur used to identify the microorganisms that were killing silkworms. In front of the microscope are some of the silkworm cocoons he studied. His research saved the silk industry in France.

“I am on the edge of mysteries and the veil is getting thinner and thinner.”

Louis Pasteur



Angel of mercy

A cartoon from a French newspaper shows Pasteur as an angel of mercy injecting a mad dog infected with rabies—a deadly killer.